

2022

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed.

Section–A

Objective type Questions

1. Choose the correct alternative for each of the following questions : $2 \times 10 = 20$

(a) In complexity theory, which case does not exist ?

- (i) Worst case
 - (ii) Best case
 - (iii) Null case
 - (iv) Average case
- (b) The graph in which, there is a closed trail which includes every edge of the graph is known as ?
- (i) Hamiltonian graph
 - (ii) Euler graph
 - (iii) Planner graph
 - (iv) Directed graph
- (c) A n -vertex graph has _____ edges.
- (i) n^2
 - (ii) $n*n$
 - (iii) $n-1$

(iv) $n * \frac{(n+1)}{2}$

(d) The value of $\left[\frac{1}{2} \cdot \left[\frac{5}{2} \right] \right]$ is ____

(i) 1

(ii) 2

(iii) 3

(iv) 0.5

(e) Power set of empty set has exactly ____ subset.

(i) One

(ii) Two

(iii) Zero

(iv) Three

(f) Find the number m of seven letters of

the words that can be formed using the letters of the word "BENZENE"

(i) 420

(ii) 530

(iii) 630

(iv) 460

(g) The relation

$\{(1,2), (1,3), (3,1), (1,1), (3,3), (3,2), (1,4), (4,2), (3,4)\}$

is :

(i) Reflexive

(ii) Transitive

(iii) Symmetric

(iv) Asymmetric

(h) The big O notation for

$f(n) = (n \log n + n^2)(n^3 + 2)$ is

(i) $O(n^2)$

(ii) $O(n^3)$

(iii) $O(n^4)$

(iv) $O(n^5)$

(i) A function is said to be ___ if and only if

$f(a) = f(b)$ implies that $a=b$ for all a and b in the domain of f .

(i) One to many

(ii) One to one

(iii) Many to many

(iv) Many to one

(j) Minimum number of unique colors required for vertex coloring of a graph is called ?

(i) Vertex matching

(ii) Chromatic Index

(iii) Chromatic number

(iv) Color number

Section-B

Short Answer type Questions

Answer any **four** questions of the following :

$$4 \times 5 = 20$$

2. Show that $f(x) = x^2 + 2x + 1$ is $O(x^2)$.

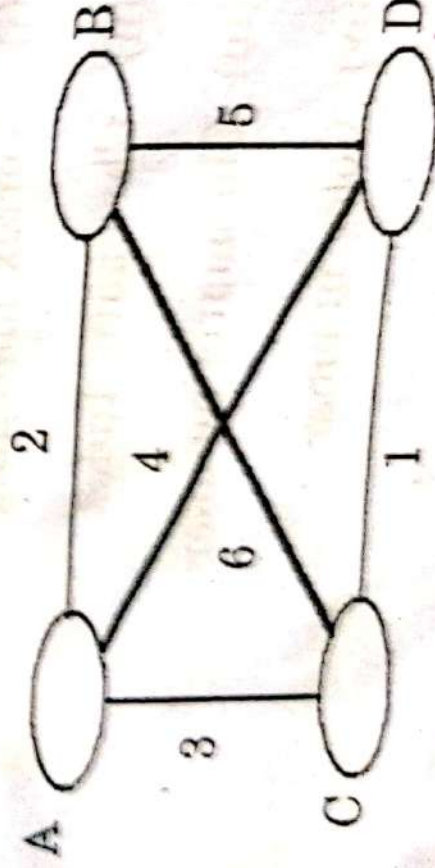
3. Evaluate the sum $\sum_{k=1}^8 (5k^2 + 8k + 1)$.

4. Use the laws of logic to show the following

logical expression is a tautology

$$((p \rightarrow q) \wedge p) \rightarrow q$$

5. What is the spanning tree ? Find the minimum spanning tree of the given weighted graph G.



6. A farmer buys 3 cows, 2 pigs and 4 horse from a man who has 6 cows, 5 pigs and 8 horse. Find the number m of choices that farmer has.

7. Use the substitution method solve the recurrence relation

$$T(n) = T(n-1) + n$$

$$T(1) = 1$$

8. Solve the recurrence relation

$$F_n = 10F_{n-1} - 25F_{n-2} \text{ whose } F_0 = 3 \text{ and } F_1 = 17$$

Section-C

Long Answer type Questions

Answer any **two** questions of the following :

$$15 \times 2 = 30$$

9. Solve the recurrence relation

$$F_n = 5F_{n-1} - 6F_{n-2} \text{ whose } F_0 = 1 \text{ and } F_1 = 4$$

10. (a) Use the laws of logic to show that

$$p \leftrightarrow q \text{ and } (p \rightarrow q) \wedge (q \rightarrow p) \text{ are}$$

equivalent.

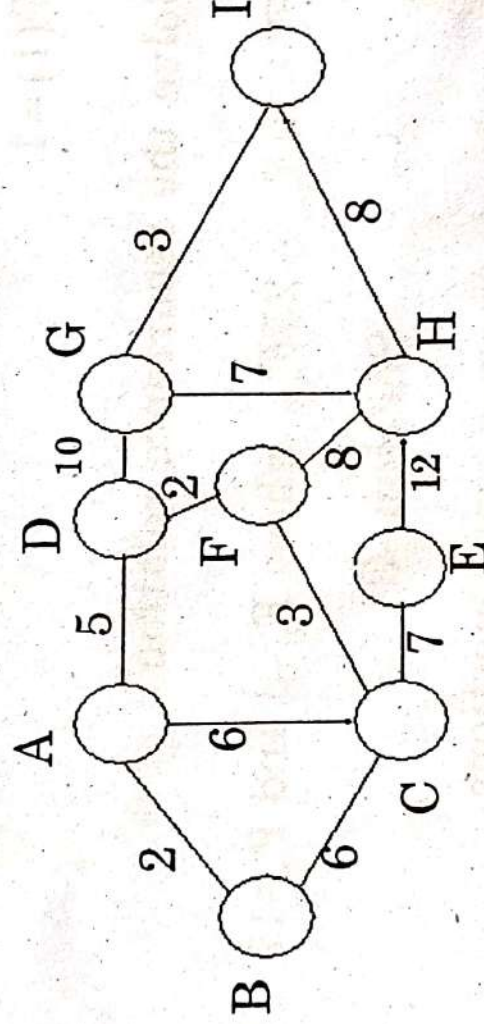
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(7)

(Turn over)

- (b) Find the minimum no of students in a class to be sure that 4 of them were born in same month.

11. Write prim's algorithm. Find the minimum cost spanning tree of the graph given below using Prim's algorithm.



12. Solve the recurrence relation

$$F_n = 3F_{n-1} + 10F_{n-2} + 7.5^n$$

where $F_0=4$ and $F_1=3$

_____ x _____